

CoRoT-2b

R-0047

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_250822 · created 2026-07-09T05:14:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460909.7362 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.224 +/- 0.065
Transit depth (Rp/Rs)^2	5.03 +/- 2.93 [%]
Orbital Inclination (inc)	89.4 +/- 2.02
Airmass coefficient 1 (a1)	4.89 +/- 3.27
Airmass coefficient 2 (a2)	-1.17 +/- 1.1
Scatter in the residuals of the lightcurve fit is	62.53 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	4.0
Transit Duration (day)	0.0996 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.224 ± 0.065 vs 0.1667 (deviation 0.87σ)
Duration fit vs published	0.0996 d vs 0.095 d (deviation 0.67σ)
Mid-time O–C	10.4 min
Depth SNR	3.4
Residual scatter	62.53 %

Light curve

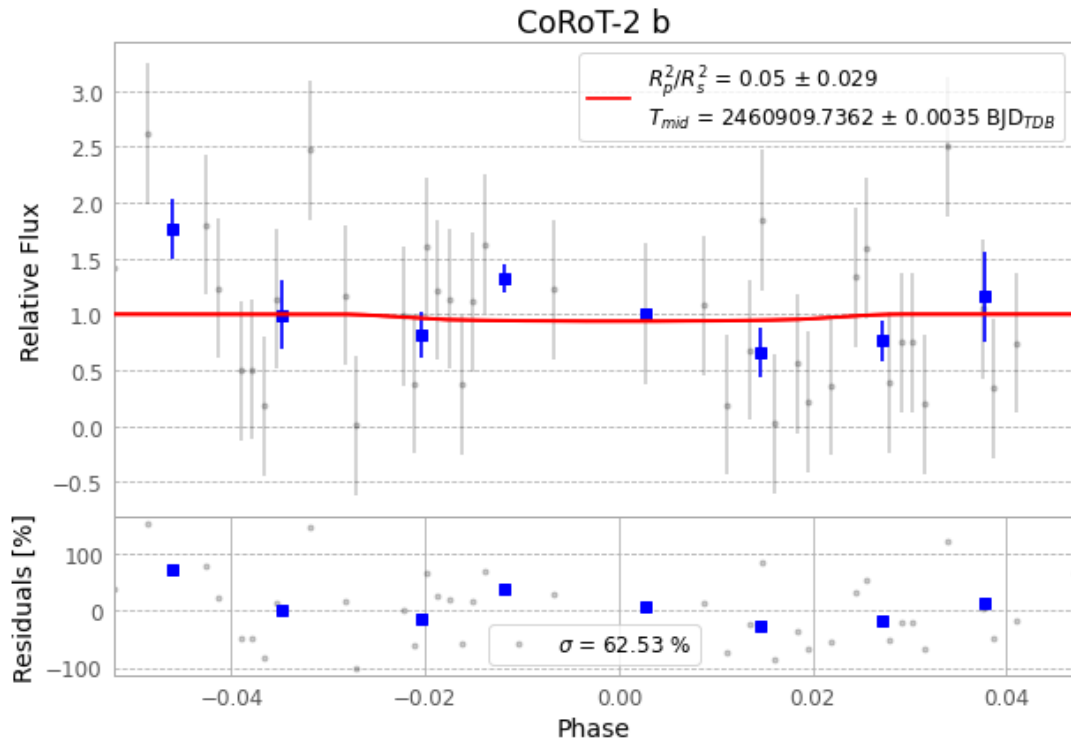


Figure 1 — FinalLightCurve_CoRoT-2 b_2025-08-21.png (SHA-256 949e2cdb5dbc855f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 49c2d8a20a6cccb2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

76 FITS frames (50 MB), CoRoT-2250822032116.FITS → CoRoT-2250822073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-250822.log (d0f587fc6e2e...), FinalLightCurve_CoRoT-2 b_2025-08-21.png (949e2cdb5dbc...), FinalLightCurve_CoRoT-2 b_2025-08-21.csv (7e80887c440e...)

Compute resources

Wall time 151.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.